

Incidence structure and solver limits of contact Delassus operators

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Abstract

The Delassus operator couples contact impulses through the bodies they share. We express this coupling as a bodywise incidence assembly with inverse spatial inertia as its block conductance. In common patch coordinates, a unique contact per body pair yields a grounded block Laplacian precisely when dynamic bodies have at most two incident patches with opposite signs. Branching violates that sign condition without invalidating the assembly or its spectral correspondence with a vertex pencil. Patch wrenches remove internal corner-force self-stress and preserve the frictionless normal and moment cone; a tangential spin ellipsoid changes the contact law. These distinctions identify where graph structure gives an exact linear solve and where it does not predict nonlinear convergence. Chain elimination reproduces dense contact updates, but a fixed spectral penalty fails on most scenes in a parameter sweep. Residual balancing lowers the median paired ratio of capped iteration counts under a common stopping rule, while losing on some scenes and incurring additional factorizations. The resulting description concerns contact conditioning and linear structure, rather than a universal solver or a spectral warning of physical failure.

1. Introduction

Repeated contact calculations enter both trajectory optimization and differentiation of robot dynamics. Contact constraints can be optimized together with a trajectory [1]. Analytical dynamics derivatives exploit the algebraic structure of the equations [2]. In an impulse formulation, eliminating body velocities leaves the Delassus operator [3], which maps contact impulses to changes in contact velocity (Le Lidec et al. [4]). A light body between heavy neighbours can make this map difficult for local iterations even in a simple stack. The operator describes this coupling, but convergence also depends on the contact law and its discretization.

For fixed geometry, we consider

$$\begin{aligned} u &= G\lambda + b, & G &= JM^{-1}J^T, \\ K_\mu \ni \lambda &\perp u + \Gamma(u) \in K_\mu^*, \\ K_\mu &= \bigcap_c \{ (\lambda_n, \lambda_t) : \lambda_n \geq 0, \|\lambda_t\| \leq \mu_c \lambda_n \}, \\ \Gamma_c(u_c) &= (\mu_c \|u_{\{t, c\}}\|, 0, 0). \end{aligned}$$

Here b is the free contact velocity including the prescribed bias, and Γ is the de Saxcé correction (de Saxcé and Feng [5], [6]; Acary et al. [7]; Cadoux [8], [9]). Dropping this correction gives an associated convex problem. We use *exact Coulomb* to distinguish the corrected contact law from that relaxation; the term does not imply exact arithmetic or convergence of every iteration. A diagonal compliance is stated separately because it changes the problem, whereas the ADMM penalty changes how a fixed problem is solved.

The contact graph has bodies as vertices and contacts as edges. Its line graph has contacts as vertices, with connections generated by shared bodies. For point masses with parallel normals, inverse mass supplies a scalar conductance. Rigid rotation requires a block conductance. A face represented by corner impulses introduces a further issue: different corner forces can produce the same body wrench. We characterize the incidence signs in common coordinates and then distinguish elimination of those redundant force coordinates from approximation of their admissible cone.

We characterize when this bodywise assembly is a block Laplacian and test whether its spectrum predicts an ADMM penalty. The Gram-matrix spectral identity is standard linear algebra, and chain elimination is a block factorization related to the Schur complements used in articulated-body dynamics (Featherstone [10], [11]). Neither identity supplies an ADMM parameter law. The central numerical comparison therefore holds the contact operator and stopping residual fixed while changing penalty rules. Separate linear-system and external-solver experiments test the corresponding algebra and formulation; their timings are not pooled into an overall performance claim. This positioning also distinguishes the experiments from the complete compliant-to-rigid solver of Carpentier and colleagues [12], which is not evaluated here.

2. Incidence and patch coordinates

2.1 Bodywise assembly

Independent body inertias let us separate the contribution of each shared body. Let M be block diagonal with positive definite blocks for dynamic bodies, and omit static bodies from its inverse. Let P_k project the global velocity onto body coordinates, and let s_k collect the incidence signs at body k . At contact c , write the Jacobian block as a signed sum of the kinematic maps of the incident bodies. The signs encode the chosen ordering of the body pair; the kinematic maps encode frames, moment arms, and the measured velocity components. Two contacts interact only through bodies incident to both of them. Expanding the mass inverse therefore gives the following assembly identity.

Lemma 1 (bodywise assembly). With contact-body signs $\sigma_{\{ck\}}$, kinematic maps $B_{\{ck\}}$, and dynamic body inertias M_k ,

$$G_{\{cc'\}} = \sum_k \sigma_{\{ck\}} \sigma_{\{c'k\}} B_{\{ck\}} M_k^{-1} B_{\{c'k\}}^T.$$

Proof. Substitute $J_c = \sum_k \sigma_{\{ck\}} B_{\{ck\}} P_k$ into $J_c M^{-1} J_{c'}^T$. The block diagonal inverse gives

$$P_k M_k^{-1} P_k^T = \delta_{\{kl\}} M_k^{-1}.$$

All terms involving distinct bodies therefore vanish. The remaining terms are precisely those in the displayed sum. If distinct contacts share at most one body, their off-diagonal coupling contains at most one such term. ■

For patch twists, all incident kinematic maps can be transported to a common world point and frame. In these coordinates the map associated with a body is independent of the particular patch, and its inverse spatial inertia becomes the conductance W_k . The body conductance remains positive definite under this transport because the transport is invertible. Here Ad_k transports a body twist to the chosen common point and frame. Each body contributes an incidence outer product multiplied by a positive definite block. This factorization explains the sparsity and positive semidefiniteness of the contact operator for arbitrary body degree. It does not, by itself, establish the signs required of a Laplacian.

$$\begin{aligned} W_k &= Ad_k M_k^{-1} Ad_k^T, \\ G_{\text{common}} &= \sum_k (s_k s_k^T) \otimes W_k. \end{aligned}$$

2.2 When the assembly is a Laplacian

The sign condition can be stated without referring to a solver. Assume one patch per body pair and use the common coordinates of the preceding subsection. If a dynamic body touches two patches with opposite signs, it contributes a negative conductance to their off-diagonal coupling. A body touching one patch contributes only to that patch's diagonal, giving a Dirichlet term. With these incidences, the operator is a block Laplacian with a possibly zero Dirichlet term. The theorem also identifies exactly why a branching contact graph falls outside this class.

Theorem 2 (Laplacian characterization). Under the common-gauge and unique-body-pair hypotheses, all off-diagonal blocks are negative semidefinite if and only if each dynamic body has degree at most two and its two incidences, when present, are opposite. Under these conditions, $G=L_H$, where shared bodies supply the block weights and singly incident bodies supply Dirichlet terms. Without these conditions, the signed assembly and positive semidefiniteness remain valid.

Proof sketch. By Lemma 1, an adjacent off-diagonal block is either $+W_k$ or $-W_k$. Since W_k is positive definite, the negative sign is necessary and sufficient. At a body with at least three incidences, two signs must coincide, producing a positive block. When the stated degree condition holds, each incidence outer product is either an edge difference outer product or a diagonal coordinate outer product. Summing these terms gives the Laplacian and its Dirichlet residue. ■

The obstruction cannot be removed by changing contact orientation. Multiplying contact rows by signs changes the incidence signs but leaves the same impossibility of pairwise opposition at a branching body. Paths and cycles admit the required orientation, including cycles of odd length. This result concerns dynamic body degree; adding a ground contact can change that degree even when the visible arrangement still resembles a ring. A ground-free ring can pass the block sign test, whereas adding a ground contact at a body already incident to a pair of patches creates the branching obstruction. Appendix A checks the assembly, rank, and spectral identities on chains and a lattice.

The scalar specialization has additional geometric restrictions. With translation only and parallel normals, the conductance is inverse mass and the kinematic maps reduce to signed scalar incidence. If the normals differ, the off-diagonal also contains their dot product. Its sign can then change even when the incidence signs are opposite, so the scalar angle condition must accompany the graph condition. Rotation adds a moment-arm term of unrestricted scalar sign. The common-frame block representation absorbs that term into spatial inertia, which is why rotation invalidates the scalar reading without invalidating the block assembly.

Friction rows provide a different test of the same reasoning. For point masses with a common orthonormal contact frame, normal and tangential coordinates produce independent copies of the scalar operator. Let E_c have the contact-frame basis vectors as rows, and let L denote the scalar signed incidence Gram matrix. When contact frames differ, their relative rotation appears between the blocks. In either case, changing the friction coefficient changes the cone while leaving this linear kinematic assembly unchanged. A frictional projected iteration can therefore fail even when the operator has the stated graph structure. The source of that failure must be sought in the interaction between projection and the linear metric.

$$\begin{aligned} G_{\{cc'\}} &= L_{\{cc'\}} E_c E_{c'}^T, \\ E_c &\equiv E \Rightarrow G = L \otimes I_3. \end{aligned}$$

2.3 Coordinates and the spectrum

The common coordinates used to identify the Laplacian need not be the coordinates used to project onto the contact cone. A block diagonal invertible map converts common patch twists to local normalized coordinates, and the Delassus operator transforms by congruence. Congruence preserves rank and positive semidefiniteness, but it does not generally preserve eigenvalues. It also does not preserve the sign of each off-diagonal block when different blocks use different transformations. Consequently, the common-frame sign theorem cannot be applied directly to a local matrix without carrying the coordinate maps. Any spectral estimate used by a solver must refer to that solver's actual Jacobian.

The spectral correspondence itself requires fewer assumptions than the Laplacian characterization. For the chosen coordinates, define

$$K = JM^{-1/2}.$$

The contact operator and the vertex operator are the two Gram matrices of this same rectangular map. Their nonzero eigenvalues coincide even on a lattice whose incidence signs violate the Laplacian condition. Thus the graph viewpoint remains useful for spectral computation outside the unsigned class. Appendix A includes the lattice in the spectral check.

Theorem 3 (vertex pencil). For any J and positive definite M , the following operators have the same nonzero spectrum with multiplicity:

$$JM^{-1}J^T, \quad M^{-1/2}J^TJM^{-1/2}.$$

Equivalently, the nonzero generalized eigenvalues of $J^TJx = \lambda Mx$ are the nonzero eigenvalues of G . Moreover, $\text{rank } G = \text{rank } J$.

Proof sketch. A nonzero eigenvector of KK^T maps through K^T to an eigenvector of K^TK with the same eigenvalue. The map is injective on that eigenspace, and the reverse map through K gives equal multiplicities. Substitution by the inverse mass square root gives the generalized pencil. Finally, multiplication by an invertible mass square root preserves rank, and a matrix and its Gram matrix have the same rank. ■

The pencil can be applied without forming the contact matrix, using products with the Jacobian, its transpose, and the inverse mass or a mass solve. This avoids contact-matrix assembly, but does not guarantee that a fixed eigensolver budget resolves the smallest positive mode. The penalty experiment uses the spectrum in the coordinates of its own contact solve.

2.4 What patch lumping preserves

Let the rectangular patch have a right-handed orthonormal frame and positive half-extents, defined by

$$(t_1, t_2, n), \quad n = t_1 \times t_2, \quad e_1 > 0, \quad e_2 > 0.$$

Normalize physical moments by their corresponding extent scales to obtain the wrench coordinates:

$$\begin{aligned} \ell_{\text{patch}} &= \sqrt{(e_1^2 + e_2^2)/3}, \\ w &= (F_n, M_{t1}/e_2, M_{t2}/e_1, F_{t1}, F_{t2}, M_n/\ell_{\text{patch}}). \end{aligned}$$

The corresponding twist coordinates are scaled dually to preserve virtual work. In the cone formulas below, moment symbols denote normalized coordinates. Summing the corner forces and their moments about the patch centroid defines the wrench map C . Its kernel consists of corner-impulse distributions with zero resultant force and zero resultant moment. Quotienting by this kernel removes redundant coordinates while preserving the generalized body impulse.

Lemma 4 (wrench quotient). For a nondegenerate rectangular patch on a free rigid body, the point Jacobian has rank six. It has twelve rows. The wrench map has rank six and satisfies

$$\begin{aligned} J_{\text{points}}^T \lambda &= J_{\text{patch}}^T C \lambda, \\ \ker C &= \ker J_{\text{points}}^T. \end{aligned}$$

Proof sketch. Differences of corner rows generate pure moments about each spatial axis, while the original rows span the resultant force directions. Their row space is therefore the full spatial wrench space. The patch Jacobian block is invertible because its force and moment directions form a basis with nonzero extent scalings. Summing corner forces and moments gives the displayed identity, and injectivity of the transposed patch block identifies the kernels. ■

The quotient also preserves a specific part of the feasible set. For frictionless normal corner forces, the centre of pressure is a convex combination of the corner positions. Conversely, any centre of pressure in the rectangle can be realized by nonnegative corner loads with the prescribed normal resultant. This proves equality of the normal and moment cone image, rather than an inclusion obtained by relaxation. The same construction extends to planar polygons by replacing the rectangle with the convex hull of the vertices. Non-collinearity is needed to retain the full wrench rank when tangential directions are included.

Theorem 5 (normal cone image). In normalized rectangular patch coordinates, the image of nonnegative normal corner impulses is

$$K_{nm} = \{(F, M_1, M_2) : F \geq 0, |M_1| \leq F, |M_2| \leq F\}.$$

Proof sketch. Write a corner contribution and the proposed corner loads as

$$\lambda(1, \alpha, \beta), \quad \alpha, \beta \in \{-1, +1\}, \\ \lambda_{\{\alpha\beta\}} = (F + \alpha M_1)(F + \beta M_2) / (4F), \quad F > 0.$$

Nonnegativity gives both moment bounds by the triangle inequality. Conversely, the displayed loads are nonnegative and recover the prescribed resultant and moments. For zero resultant, the inequalities force zero moments. In physical coordinates these are precisely the centre-of-pressure inequalities. ■

This exact statement does not extend to the tangential cone used in the patch solver. The linear map retains normal torque, but the chosen Contensou-type ellipsoid has a different spin capacity from the corner Coulomb cones [13]. Pure spin isolates the difference because tangential corner forces can form a torque without a tangential resultant. For the same normal impulse, the capacities are

$$\tau_{\text{corner}} = \mu F_n / (e_1^2 + e_2^2), \\ \tau_{\text{patch}} = \mu F_n l_{\text{patch}}, \\ \tau_{\text{corner}} / \tau_{\text{patch}} = \sqrt{3}.$$

Pure spin therefore distinguishes a change of contact law from loss of a wrench component.

Local rank restoration does not imply full row rank of the scene Jacobian. Loops can sustain wrench self-stress after corner redundancy is removed, as the lattice in Appendix A demonstrates. If the complete patch Jacobian has full row rank, positive definiteness makes the fixed-shift affine variational inequality coercive and strongly monotone on the closed convex cone. It therefore has a unique solution. A locally constant active set, strict complementarity, and a nonsingular active-set residual Jacobian permit application of the implicit function theorem. These hypotheses neither establish uniqueness of the full Coulomb fixed point nor differentiation through a contact-regime change.

3. Linear consequences

On a patch chain, only adjacent interfaces share a dynamic body. The Delassus matrix is therefore block tridiagonal in chain order, and adding an ADMM diagonal shift preserves that pattern. Eliminating successive diagonal blocks solves the same linear update as a dense factorization. Appendix B compares the resulting outer iterations. The converged cases reproduce the dense updates; the driven mass-ratio column fails with both factorizations. Agreement of the linear solves does not make the outer contact map contractive.

This observation should not be transferred directly to projected iteration. Preconditioning changes the metric of the linear step. When a cone constraint is active, a Euclidean projection need not be the projection associated with that metric. An exact inverse can consequently solve an interior linear problem without yielding a valid single-step constrained update. ADMM

separates the linear solve from cone projection, so the block factorization has an unambiguous role there.

Classical algebraic multigrid provides a separate linear baseline. The operator in Appendix C is the scalar incidence Gram matrix with inverse-mass weights, shifted when necessary to obtain a positive definite system. It is the normal Delassus operator only under the translation and frame assumptions stated above. Both multigrid variants reach the linear tolerance with conjugate-gradient acceleration. Their standalone cycles stall on the tested non-M-matrices, while the chain cases converge. This coincidence does not establish a causal consequence of the sign theorem. The experiment evaluates classical multigrid, not the nearly-linear algorithms of Spielman-Teng, Koutis-Miller-Peng, or Kyng-Sachdeva (Spielman and Teng [14]; Koutis et al. [15]; Kyng and Sachdeva [16]).

A graph surrogate can also be a poor preconditioner for the actual contact matrix. Appendix D compares the graph, block-Jacobi, and unpreconditioned methods on identical right-hand sides within each scene. The graph construction agrees with the relevant normal block on the mass-ratio chain and reduces Krylov work. On the packing it differs from that block and costs more iterations and more solve time than Jacobi scaling. Those times exclude preconditioner construction, whereas the multigrid times include setup. These cases show a benefit when the surrogate agrees with the tested block and a loss when it does not. They do not isolate operator agreement as the cause of that difference.

4. Penalty selection for a fixed contact problem

The incidence identity permits spectral computation through Jacobian products without assembling the contact matrix. It does not guarantee a prescribed eigenvalue accuracy at a fixed Krylov budget. More fundamentally, accurate eigenvalues do not contain the contact activity or projection derivative that governs a nonlinear iteration. We test the resulting parameter prediction with a dense spectrum so that estimation error cannot explain its failures.

4.1 Sweep and paired comparison

Appendix E uses the same operator for the spectrum, ADMM update, de Saxcé correction, and stopping residual. Write that operator as

$$\begin{aligned} G_\eta &= G_\theta + \text{diag}(\eta), & G_\theta &= JM^{\{-1\}}J^T, \\ R_\eta(\lambda) &= \|\lambda - \Pi_K[\lambda - D(G_\eta\lambda + b + \Gamma(G_\eta\lambda + b))]\|_\infty, \\ D_c &= \|(G_\eta)_{\{cc\}}\|_2^{\{-1\}} I. \end{aligned}$$

The packing and granular-step cases include diagonal compliance; its multiplier is included in the table. The remaining scenes use the rigid operator. Thus the packing spectrum in this experiment is not the unregularized spectrum in the preconditioner comparison. Each penalty candidate starts from zero. The reference penalty is the converged grid point requiring the fewest iterations, with ties resolved toward the smaller penalty. Boundary winners are grid minima, not established optima beyond the sampled range.

The candidate rules and grid are

$$\begin{aligned} \rho_A &= \lambda_{\min}^+(G_\eta), & \rho_B &= \sqrt{(\lambda_{\min}^+(G_\eta)\lambda_{\max}(G_\eta))}, \\ \rho_{\text{struct}} &= \rho_A \text{ if the structural test passes, and } \rho_B \text{ otherwise,} \\ \rho_j &= \lambda_{\min}^+(G_\eta) 10^{\{a_j\}}, & a_j &\in \text{linspace}(a_{\min}, a_{\max}, N_{\text{grid}}). \end{aligned}$$

The structural test checks the Jacobian support: each represented body has at most two incident contacts, distinct contacts share at most one body, and the Jacobian blocks of paired incidences are opposite. Its result depends on the represented coordinates; multiple corner contacts count separately before patch reduction. Articulated cases use their generalized operator in the sweep

and are not evidence for a bodywise Laplacian decomposition of a coupled mass matrix. Appendix E states the eigenvalue cutoff, grid range, and failure counts alongside each scene.

The structural rule misses the factor-of-two parameter criterion on most scenes. Chains with inactive constraints often favour a boundary grid value, while packings prefer values above the geometric-mean scale. The same spectral choice is therefore not uniformly predictive across regimes. Selecting whichever spectral rule lies nearer the grid winner after the sweep would change the task from prediction to retrospective selection.

Appendix F is the primary algorithmic comparison. It reruns the swept penalty at the same natural-map target used for every fixed and adaptive method in that table. The methods share zero initialization and the ADMM/Cadoux implementation with sparse direct linear solves. The mean initialization is $\text{tr}(G_\eta)/\dim(G_\eta)$. The structural initialization is the declared rule above. The He-Yang-Wang residual-balancing rule increases or decreases the penalty when primal and dual residual norms become imbalanced (Boyd et al. [17]; He et al. [18]). The normalized variant applies the same decision to relative residuals, following Wohlberg [19]. The implemented residuals are

$$\begin{aligned} r_p &= x - z, & r_d &= p(z - z_{\text{previous}}), \\ r_p, \text{rel} &= \|r_p\|_2 / \max(\|x\|_2, \|z\|_2, \epsilon_{\text{den}}), \\ r_d, \text{rel} &= \|r_d\|_2 / \max(\|y\|_2, \epsilon_{\text{den}}), & \epsilon_{\text{den}} &= 10^{-300}. \end{aligned}$$

Here y is the unscaled dual variable and remains unchanged when the penalty changes. Every penalty change triggers a new sparse factorization, so iteration counts do not measure equal amounts of work.

The measured solver freezes the de Saxcé shift during each inner ADMM block. It starts with zero shift and refreshes tangential speeds after the inner primal residual meets a numerical threshold. The threshold depends on the change in the shift, with the free-velocity bias used for the first block. Appendix F specifies this schedule. The local proposition below instead refreshes the correction at every iteration.

He-Yang-Wang adaptation from either initialization reduces the median of the recorded scene-wise iteration ratios against the fixed structural setting. The opening column nevertheless requires more iterations with either initialization, and every adaptive method is capped on the dense random system although the fixed rule converges. Raw iteration medians order the initializations differently from paired ratios, and the normalized structural variant has the largest convergence count. The data therefore support residual adaptation within this implementation without establishing a preferred initialization independently of the evaluation metric. A run stopped at its cap remains a failure to attain the target. Its count is a lower bound on its own convergence work, but a ratio of two capped counts is not in general a lower bound on their convergence-time ratio.

4.2 A local obstruction to a spectral rule

A sliding contact gives a direct explanation for the missing information. The correction depends on tangential velocity even if the Delassus matrix is isotropic. Consequently, the local iteration depends on friction and on the schedule used to refresh that correction.

Proposition 6 (local sliding rate).

Let $G = gI$ with positive g , use Euclidean cone projection without over-relaxation, and refresh $\Gamma(Gz + b)$ every iteration. Assume positive friction and penalty parameters. At a fixed point, write

$$\lambda = \ell(1, \mu e), \quad u = (\theta, -ve), \quad \|e\| = 1.$$

The vector e is tangent to the contact plane. Here ℓ is the positive normal impulse and v is the positive slip speed. The linearized spectrum is

$$\begin{aligned} &\{0, 0, 0, g/(g+\rho), \\ &\quad (\rho+g\mu^2/(1+\mu^2))/(g+\rho), \\ &\quad (\rho^2+gk)/((g+\rho)(\rho+k))\}, \quad k=v/(\mu\ell). \\ &\rho_{\text{opt}} = g/(1+\mu^2), \quad q_{\text{opt}} = (1+\mu^2)/(2+\mu^2). \end{aligned}$$

Proof. Write the scaled dual variable as d , and denote the de Saxcé correction by S . The differentiated map is

$$\begin{aligned} A &= (G+\rho I)^{-1}, \\ x^+ &= A[\rho(z-d) - b - S(Gz+b)], \\ w &= x^+ + d, \quad z^+ = \Pi_K \mu(w), \quad d^+ = w - z^+. \end{aligned}$$

In the intermediate projection branch, define

$$\begin{aligned} w &= (a, re), \quad \ell = (a+\mu r)/(1+\mu^2), \\ q &= (1, \mu e)/\sqrt{1+\mu^2}, \quad c = (0, e_{\text{perp}}). \end{aligned}$$

The transverse tangent is a unit vector orthogonal to e . Differentiation gives

$$\begin{aligned} P' &= qq^T + \chi cc^T, \quad \chi = \mu\ell/r = \rho/(\rho+k), \quad k = v/(\mu\ell), \\ D &= S'(u)G, \quad B = A(\rho I - D), \quad C = I - \rho A, \\ T' \rho &= [P'; I - P'] [B, C], \quad H = BP' + C(I - P'). \end{aligned}$$

The full derivative has the eigenvalues of H together with additional zeros. Choose the normal-tangent plane coordinates with tangent axis opposite to e , and define

$$\begin{aligned} b_1 &= (1, -\mu)/\sqrt{1+\mu^2}, \quad b_2 = (\mu, 1)/\sqrt{1+\mu^2}, \\ a_\rho &= \rho/(g+\rho), \quad \beta = g/(g+\rho). \end{aligned}$$

For the isotropic operator, substitution in this basis gives the triangular plane block

$$\begin{bmatrix} a_\rho + \beta\mu^2/(1+\mu^2) & 0 \\ \beta\mu^3/(1+\mu^2) & \beta \end{bmatrix}.$$

The transverse eigenvalue is $a_\rho\chi + \beta(1-\chi)$, a convex combination of a_ρ and β . The first plane eigenvalue dominates a_ρ , so the transverse branch cannot exceed the largest plane branch. The decreasing and increasing plane branches meet at the stated penalty, giving the unique spectral-radius minimum. With the correction frozen, the derivative term D vanishes and the plane eigenvalues are a_ρ and β . Their crossing is

$$\rho = g, \quad q_{\text{frozen}} = 1/2.$$

None of these derivatives applies across zero slip speed without separate directional analysis. ■

This proposition concerns the asymptotic rate near the specified sliding fixed point. It does not minimize cold-start iterations at a finite tolerance, and its smooth differentiation does not apply across zero slip speed. Freezing the correction changes the differentiated map while leaving the Delassus operator unchanged. A spectral penalty rule that ignores the contact law and refresh schedule therefore cannot be deduced from the operator identity.

5. Contact-law checks and scope

Appendix G compares APGD and projected Gauss-Seidel for the convex relaxation with ADMM for the corrected Coulomb law. All recorded impulses pass the tangential inequality test, including the capped runs. That is compatible with projection onto the force cone and does not establish the original velocity complementarity condition. The cube shows a different normal-velocity diagnostic even when the relaxed solver reaches its own residual target. Conversely,

corrected Coulomb solutions can have positive normal velocities at separating contacts. The diagnostic is defined as

$$\text{gap} = \Delta t \max(0, \max_c u_{\{n,c\}}).$$

The step scale and force-cone test tolerance are in Appendix G. Because the statistic includes separating contacts, its maximum is not an estimate of artificial lift. The method-specific residual column and cap status must also be retained: a convex target and a corrected natural-map target are not interchangeable accuracy requirements. These measurements test the effect of the formulation and budget rather than a universal speed ordering.

Appendix H evaluates external friction-contact implementations on the same exported operators, biases, and friction coefficients. Siconos uses $u = Mr + q$, with $M = G_\eta$, $r = \lambda$, and $q = b$. The evaluated methods are nonsmooth Gauss-Seidel, the ACLM fixed point, proximal point, ADMM, de Saxcé fixed point, and nonsmooth Newton. The relative Siconos complementarity error is reported beside the absolute natural-map residual evaluated on the returned reaction. A run can pass the former test while exceeding the same numerical threshold in the latter. These are formulation checks, not matched-accuracy timing comparisons.

Different converged reactions can also arise from redundant force coordinates. The patch kernel identifies impulses invisible to body motion, but rank deficiency alone does not prove that any observed difference belongs to that kernel. Small residuals are necessary before interpreting a reaction as another admissible selection; capped, nonfinite, and missing results remain failures. The internal reference is itself a finite-tolerance solution. The stored cube and lattice references do not attain the absolute natural-map target used in the penalty-adaptation comparison. Relative reaction differences against them are therefore diagnostics, not independent error bounds on Siconos. A comparison of generalized body impulses at a common residual target is still needed to resolve force selection quantitatively.

The same distinction between operator and admissible reaction limits physical interpretation. At fixed geometry and inertia, changing load changes b without changing G . A reaction can reach the contact-cone boundary while the Delassus spectrum stays fixed. The spectrum can diagnose a difficult linear solve without locating a sliding or tipping threshold. In normalized patch coordinates and for positive supporting loads and friction, the tested cone margins take the form

$$\begin{aligned} m_1 &= \min_c (F_{\{n,c\}} - |M_{\{t1,c\}}|) / F_{\{n,c\}}, \\ m_2 &= \min_c (F_{\{n,c\}} - |M_{\{t2,c\}}|) / F_{\{n,c\}}, \\ m_f &= \min_c (\mu_c F_{\{n,c\}} - \|(F_{\{t1,c\}}, F_{\{t2,c\}}, M_{\{n,c\}})\|) / (\mu_c F_{\{n,c\}}), \\ m &= \min(m_1, m_2, m_f). \end{aligned}$$

A moment component alone is not this combined test, and the spin coordinate enters the friction norm. Normal-force positivity must be checked separately. These margins describe the chosen patch cone; they are not asserted to equal a ZMP margin or to predict a dynamic tipping event under arbitrary loading.

All retained timings are CPU measurements on a shared host. The multigrid comparison includes setup; the preconditioner comparison measures the stated batch of solves after construction. No GPU speed claim is made. Repeatability checks apply only to runs for which paired outputs exist. They do not resolve differences in contact law or stopping residual.

6. Conclusion

Inverse spatial inertia gives an exact bodywise representation of contact coupling. Its reduction to a grounded block Laplacian requires common coordinates, unique body pairs, and the incidence sign condition. Patch lumping removes internal point-force self-stress, while its admissible tangential wrench remains a constitutive choice. These properties justify chain

elimination and explain why a scalar surrogate can fail on a packing. They do not select a universal nonlinear penalty: adaptation reduces the median paired ratio of recorded counts but can fail where the fixed setting converges. Graph structure is consequently a tool for specifying and solving the linear part of contact, whose usefulness must be assessed together with projection, compliance, and the stopping residual.

Figures

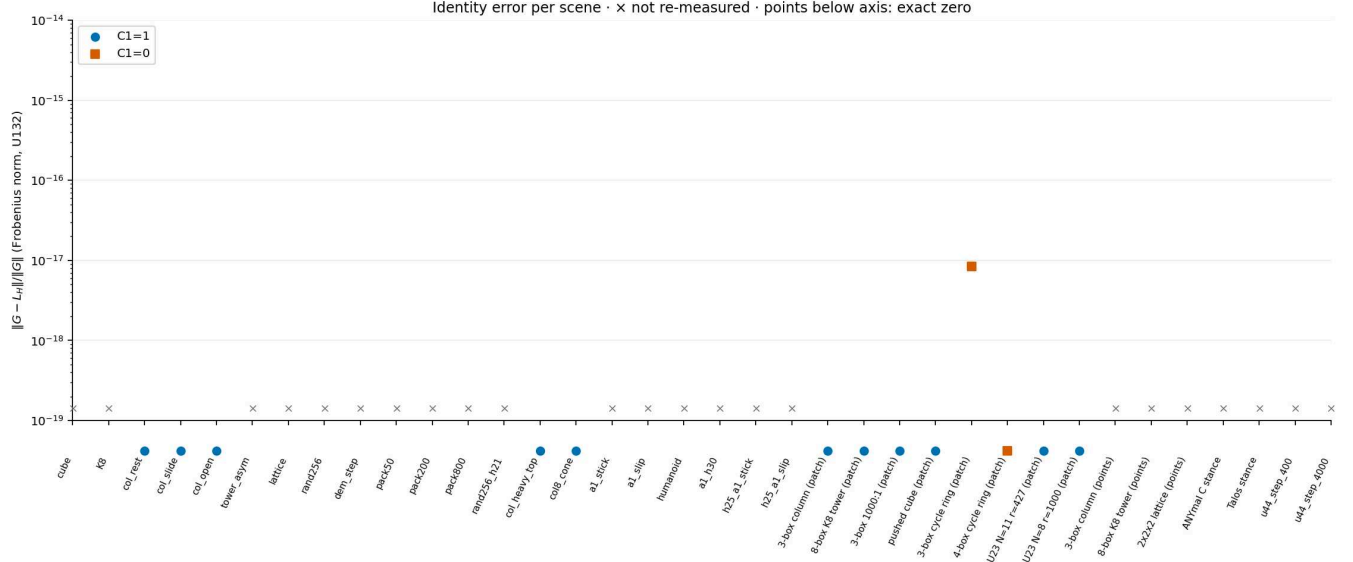


Figure 1 — identity error by scene and C1 status. Thirteen rows contain measured values of $\|G - L_H\|_F / \|G\|_F$. Twenty-three rows have no L_H for their J and are shown as crosses. $C1$ is computed from J . Twelve zero values are marked below the logarithmic axis. Factorization error and the normal-block surrogate are excluded.

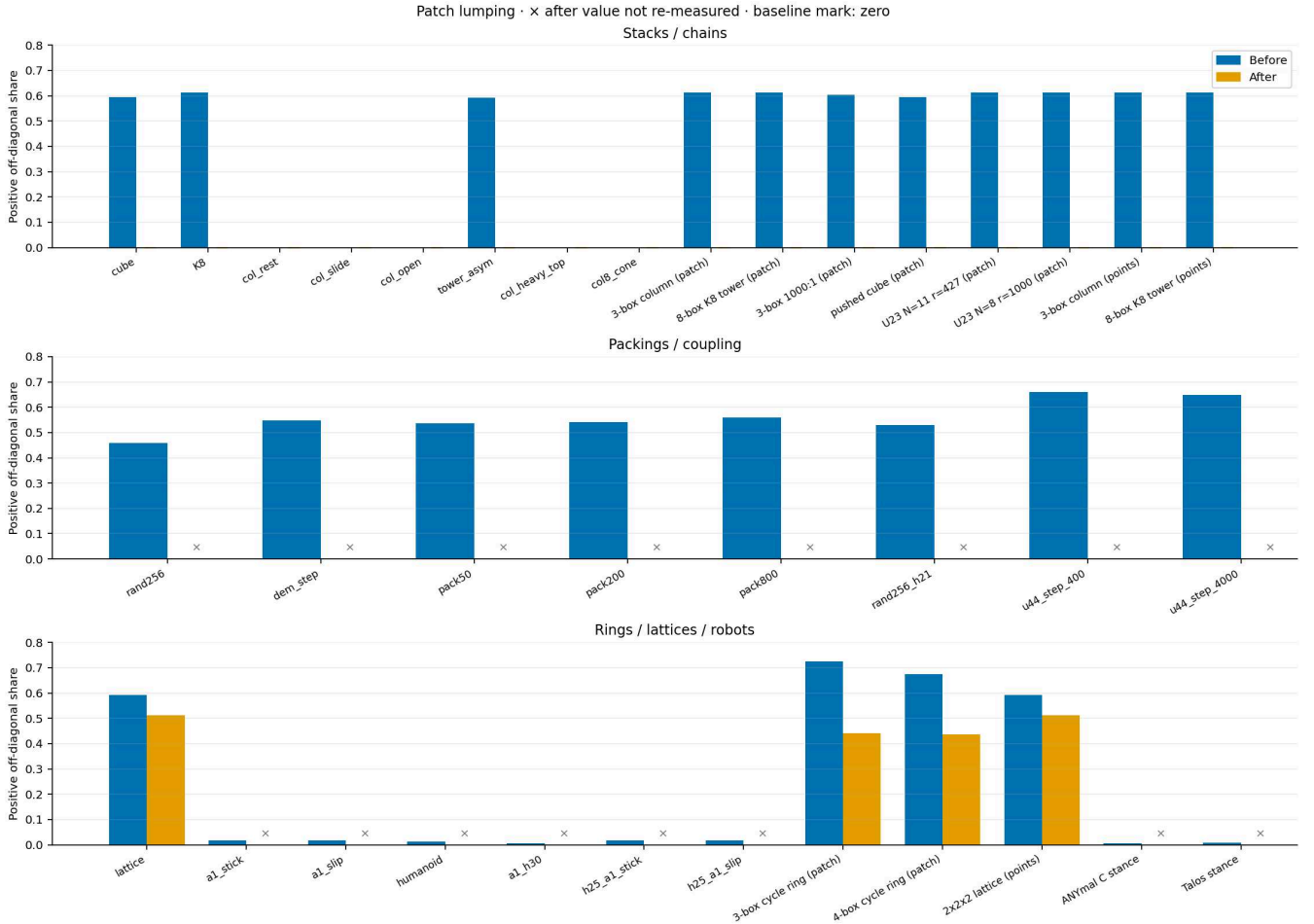


Figure 2 — positive off-diagonal Frobenius share. Before values cover 36 scenes. Sixteen after values lack a defined transform or have an operator mismatch and are shown as crosses. Patch-row before values use the point model; structure and after values use the patch model. The robot net-wrench diagnostic is excluded.

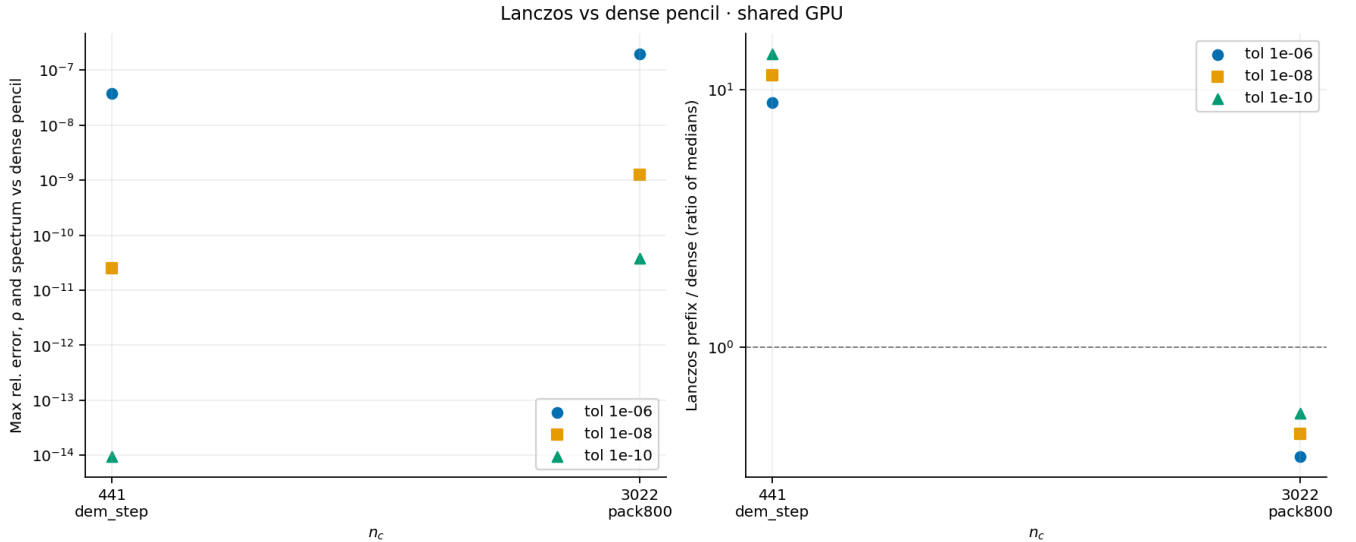


Figure 3 — pencil-spectrum error and cost against a dense eigensolve. The panels show the recorded Lanczos prefix and error for the granular step and the 800-grain packing. Error is the maximum relative error in ρ , $\lambda_{\min}^+$ and $\lambda_{\max}$. The timing ratio divides medians and excludes the reduced eigensolve and result extraction. GPU shared applies to these source timings.

Appendix A. Assembly, rank, and spectral correspondence

The lattice retains the assembly and pencil identity while patch reduction leaves global rank deficiency.

Scene	Patches [count]	Bodies of degree >2 [count]	Point rank/rows [count]	Patch rank/rows [count]	Assembly relative ∞ -error [1]	Pencil relative spectral error [1]
Three-box column	3	0	18/36	18/18	2.368e−17	6.510e−16
Eight-box tower	8	0	48/96	48/48	9.586e−17	9.565e−16
Mass-ratio column (1000:1)	3	0	18/36	18/18	2.368e−17	4.836e−16
Box lattice (2×2×2)	16	8	48/192	48/96	2.368e−17	1.115e−15

Appendix B. Dense and block elimination

The final row compares unconverged iterates and does not measure solution error.

Scene	Dense updates [count]	Block updates [count]	max abs impulse difference [N s]	Status	Natural-map target [N s]
Eight-box tower, gravity	140	140	1.67e−16	converged	1e−8
Eight-box tower, rolling	1848	1848	2.37e−15	converged	1e−8
Mass-ratio column (1000:1), gravity	296	296	4.39e−12	converged	1e−8
Mass-ratio column (1000:1), rolling	>30000	>30000	1.22e+01	both capped; iterate difference	1e−8

Appendix C. Scalar incidence linear systems

Setup-inclusive CPU time and attained relative residual distinguish converged linear solves from stalled standalone cycles on $A = S \text{ diag}(1/m) S^T + \rho I$.

Ruge-Stüben + CG

Conjugate-gradient acceleration reaches the target on every listed incidence system.

Scene	Dimension [count]	ρ [kg ^{−1}]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e−10	Status
Eight-box tower incidence	32	0.02	34	0.003703660099	1.39391005e−11	pass
Mass-ratio patch incidence	3	0	2	0.000184888020	1.697824553e−13	pass
Eight-body patch incidence	8	0	2	0.0001980180386	8.712091731e−15	pass
Asymmetric patch incidence	8	0	2	0.0002546079922	2.686039817e−12	pass
Mass-ratio point chain	3	0	2	0.000187835074	1.228302313e−13	pass
Eight-body point chain	8	0	2	0.0001932290616	8.712091731e−15	pass

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Granular step	441	0.358098622	93	0.02097494493	8.859897596e-11	pass
Packing, 200 grains	710	2.864788976	108	0.03051988408	6.781539484e-11	pass
Packing, 50 grains	176	2.864788976	80	0.01271233603	5.479209081e-11	pass
Packing, 800 grains	3022	2.864788976	130	0.1069332911	5.3348256e-11	pass
Asymmetric tower incidence	32	0.01001	30	0.003410444013	2.329227715e-11	pass

Ruge-Stüben V-cycle

Standalone cycles miss the target on the branching point-contact and packing cases.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	32	0.02	201	0.009372626082	0.07636040734	target missed
Mass-ratio patch incidence	3	0	2	8.222402539e-05	2.619951042e-13	pass
Eight-body patch incidence	8	0	2	8.537597023e-05	1.082399872e-14	pass
Asymmetric patch incidence	8	0	2	8.883082774e-05	3.020610497e-12	pass
Mass-ratio point chain	3	0	2	8.282402996e-05	9.445502438e-14	pass
Eight-body point chain	8	0	2	8.821592201e-05	1.082399872e-14	pass
Granular step	441	0.358098622	201	0.03159748495	0.04891895978	target missed
Packing, 200 grains	710	2.864788976	201	0.04080992204	0.05939238848	target missed
Packing, 50 grains	176	2.864788976	201	0.01927015395	0.03547710012	target missed
Packing, 800 grains	3022	2.864788976	201	0.131928546	0.06840721127	target missed
Asymmetric tower incidence	32	0.01001	201	0.009541906882	0.2012310665	target missed

Smoothed aggregation + CG

Conjugate-gradient acceleration reaches the target, with more iterations on packings than on chains.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	32	0.02	31	0.004100079997	5.511167299e-12	pass

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Mass-ratio patch incidence	3	0	2	0.0002522680443	1.697824553e-13	pass
Eight-body patch incidence	8	0	2	0.0002377199708	8.712091731e-15	pass
Asymmetric patch incidence	8	0	2	0.0002507490572	2.686039817e-12	pass
Mass-ratio point chain	3	0	2	0.0002478290116	1.228302313e-13	pass
Eight-body point chain	8	0	2	0.0002561140573	8.712091731e-15	pass
Granular step	441	0.358098622	128	0.02600767615	7.399822361e-11	pass
Packing, 200 grains	710	2.864788976	141	0.03725839104	7.544355978e-11	pass
Packing, 50 grains	176	2.864788976	103	0.01779325691	3.833834424e-11	pass
Packing, 800 grains	3022	2.864788976	160	0.1279786841	9.398701155e-11	pass
Asymmetric tower incidence	32	0.01001	32	0.004393324023	4.394401455e-12	pass

Smoothed aggregation V-cycle

The standalone cycle reaches the target on chains and stalls on the other listed systems.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	32	0.02	201	0.01025248098	0.1194322982	target missed
Mass-ratio patch incidence	3	0	2	0.0001476858743	2.619951042e-13	pass
Eight-body patch incidence	8	0	2	0.0001562670805	1.082399872e-14	pass
Asymmetric patch incidence	8	0	2	0.0001603679266	3.020610497e-12	pass
Mass-ratio point chain	3	0	2	0.0001517349156	9.445502438e-14	pass
Eight-body point chain	8	0	2	0.0001542770769	1.082399872e-14	pass
Granular step	441	0.358098622	201	0.02781967295	0.06168650859	target missed
Packing, 200 grains	710	2.864788976	201	0.03804021899	0.07377174546	target missed
Packing, 50 grains	176	2.864788976	201	0.02202152892	0.04828668837	target missed
Packing, 800 grains	3022	2.864788976	201	0.1277840671	0.0864185493	target missed
Asymmetric tower incidence	32	0.01001	201	0.01050093805	0.07075948385	target missed

Direct

Direct solves reach the target on every listed incidence system.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	32	0.02	1	7.706589531e-05	1.311443007e-14	pass
Mass-ratio patch incidence	3	0	1	1.878698822e-05	1.162189216e-13	pass
Eight-body patch incidence	8	0	1	1.735088881e-05	1.899354946e-15	pass
Asymmetric patch incidence	8	0	1	2.294604201e-05	8.703065953e-13	pass
Mass-ratio point chain	3	0	1	1.912808511e-05	1.133448696e-13	pass
Eight-body point chain	8	0	1	1.739000436e-05	1.899354946e-15	pass
Granular step	441	0.358098622	1	0.00155614491	5.148442586e-14	pass
Packing, 200 grains	710	2.864788976	1	0.00382089312	4.521111564e-14	pass
Packing, 50 grains	176	2.864788976	1	0.00019621301	2.397510841e-14	pass
Packing, 800 grains	3022	2.864788976	1	0.03917622194	1.983819478e-14	pass
Asymmetric tower incidence	32	0.01001	1	3.126298543e-05	2.731240243e-14	pass

Jacobi CG

Jacobi-preconditioned CG reaches the target without a multigrid hierarchy.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Eight-box tower incidence	32	0.02	10	0.00011357094	7.474580249e-11	pass
Mass-ratio patch incidence	3	0	3	4.09099739e-05	6.465091218e-13	pass
Eight-body patch incidence	8	0	8	8.581799921e-05	2.415574774e-15	pass
Asymmetric patch incidence	8	0	8	8.863897528e-05	6.88798532e-13	pass
Mass-ratio point chain	3	0	3	4.1479012e-05	7.79024578e-13	pass
Eight-body point chain	8	0	8	8.650007658e-05	2.415574774e-15	pass
Granular step	441	0.358098622	97	0.00128661794	6.076249474e-11	pass
Packing, 200 grains	710	2.864788976	115	0.00180110603	7.394944173e-11	pass
Packing, 50 grains	176	2.864788976	69	0.00077530101	3.160264002e-11	pass

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Packing, 800 grains	3022	2.864788976	152	0.007260465994	9.292384405e-11	pass
Asymmetric tower incidence	32	0.01001	11	0.0001225449378	9.733895466e-14	pass

Tridiagonal Cholesky

The chain systems admit tridiagonal factorization and reach the target.

Scene	Dimension [count]	ρ [kg ⁻¹]	Iterations [count]	Setup + solve [s]	Relative residual [1], target 1e-10	Status
Mass-ratio patch incidence	3	0	1	9.471084923e-06	1.162189216e-13	pass
Eight-body patch incidence	8	0	1	1.657998655e-05	1.899354946e-15	pass
Asymmetric patch incidence	8	0	1	1.723202877e-05	8.14012696e-13	pass
Mass-ratio point chain	3	0	1	9.119976312e-06	1.133448696e-13	pass
Eight-body point chain	8	0	1	1.666997559e-05	1.899354946e-15	pass

Appendix D. Preconditioned contact linear solves

Each scene uses identical right-hand sides across methods, with CPU times excluding preconditioner construction.

Graph

The graph preconditioner reduces iterations on the column but requires more iterations and solve time on the packing than the other methods.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]	Max relative residual [1], target 1e-10	max abs(L-G_nn) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	0.002420391178	1	0.002859290107	9.25577049e-14	0	100
Packing, 50 grains	469.2924618	62	2.71804603	9.807603499e-11	286.4727554	100

Block Jacobi

On the packing, block Jacobi costs less solve time than the graph preconditioner.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]	Max relative residual [1], target 1e-10	max abs(L-G_nn) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	0.002420391178	3	0.003835485084	1.002617841e-13	0	100
Packing, 50 grains	469.2924618	35	0.20877871	9.253055396e-11	286.4727554	100

None

The unpreconditioned packing solve costs less time than either tested preconditioner.

Scene	ρ [kg ⁻¹]	Median iterations [count]	Batch solve time [s]	Max relative residual [1], target 1e-10	max abs(L-G_nn) [kg ⁻¹]	Right-hand sides [count]
Mass-ratio column	0.002420391178	3	0.003077984089	6.764498941e-11	0	100
Packing, 50 grains	469.2924618	33	0.1690009242	7.108833378e-11	286.4727554	100

Appendix E. Spectral penalties and grid minima

The parameter distance is $\max(\rho/\rho_{struct}, \rho_{struct}/\rho)$ and capped or nonfinite candidates remain in the accounting.

Scene	Contacts [count]	λ_{min}^+ [kg ⁻¹]	λ_{max} [kg ⁻¹]	ρ^* [kg ⁻¹]	Iterations [count]	Rule	Parameter distance [1]	Boundary	Capped [count]	Nonfinite [count]	$\eta_c/ G0_{cc} _2$ [1]
Cube	4	0.111696312	0.9549703547	0.1986272518	58	B	1.644279068	false	6	0	0
Eight-box tower	32	0.00233314724	46.37933351	2.33314724e-06	2	B	140990.931	true	0	0	0
Mass-ratio column, rest	3	0.000292768252	2.00100025	2.92768252e-07	4	A	1000	true	3	0	0
Mass-ratio column, sliding	3	0.000292768252	2.00100025	0.0009258145028	162	A	3.16227766	false	11	0	0
Mass-ratio column, opening	3	0.000292768252	2.00100025	0.001646356868	27	A	5.623413252	false	5	0	0
Asymmetric tower	32	3.77871314e-06	24.02119446	3.77871314e-09	2	B	2521304.725	true	1	0	0
Box lattice	64	0.4021119685	65.15994714	0.0004021119685	2	B	12729.66169	true	0	0	0
Random contact network	256	0.2163405649	25.76148555	3.847139722	202	B	1.629609426	false	12	0	0
Granular step	441	0.6172496689	526.1260007	61.72496689	1104	B	3.425196964	false	22	0	0.01
Packing, 50 grains	176	4.935228954	4306.254112	493.5228954	577	B	3.385351949	false	21	0	0.01
Packing, 200 grains	710	4.870694009	4386.170694	487.0694009	833	B	3.332365726	false	16	5	0.01
Packing, 800 grains	3022	4.773906827	4759.045125	477.3906827	926	B	3.167211443	false	16	6	0.01
Dense random contact system	256	126.7338501	2889.305391	225.3681962	800	B	2.685037834	false	15	6	0

Scene	Contacts [count]	$\lambda_{\min}^+$ [kg ⁻¹]	$\lambda_{\max}$ [kg ⁻¹]	ρ^* [kg ⁻¹]	Iterations [count]	Rule	Parameter distance [1]	Boundary	Capped [count]	Nonfinite [count]	$\eta_c/ G0_{cc} _z$ [1]
Column, heavy top	3	0.000292768252	2.00100025	0.01646356868	2681	A	56.23413252	false	22	0	0
Eight-body constrained column	8	7.602904289e-05	2.001707419	0.01352008815	3027	A	177.827941	false	23	0	0
Quadruped, sticking	4	0.01683484815	5.586013882	1.683484815e-05	2	B	18215.72258	true	0	0	0
Quadruped, sliding	4	0.01683484815	5.586013882	0.946693082	627	B	3.087120605	false	15	0	0
Humanoid	2	0.05780543824	0.5933695941	5.780543824e-05	3	B	3203.89515	true	3	0	0
Quadruped, unloading probe	4	0.2721286926	7.610089619	2.721286926	56	B	1.891003279	false	8	0	0
Quadruped, batch sticking state	4	0.01898863344	5.586100305	0.1898863344	200	B	1.715171333	false	10	0	0
Quadruped, batch sliding state	4	0.01756848743	5.586491561	0.5555643533	248	B	1.773362876	false	14	0	0

$\lambda_{\min}^+ > 10^{-11} \lambda_{\max}$, $\text{quad } (a_{\min}, a_{\max}, N_{\text{grid}}) = (-3, 3, 25)$, $\text{quad } R_{\eta} < 10^{-8}$; $N \setminus s$, $\text{quad } I_{\max} = 5000$.

Appendix F. Penalty adaptation with a common residual

C marks a capped run; every method uses the same operator, cold start, and natural-map target, including the rerun grid-reference penalty.

Scene	Cap [iterations]	Grid reference [iterations]	Fixed structure [iterations]	He, mean [iterations]	He, structure [iterations]	Normalized, mean [iterations]	Normalized, structure [iterations]
Cube	30000	70	75	69	75	143	118
Eight-box tower	30000	2	48	13	9	13	9
Mass-ratio column, rest	30000	5	42	21	10	21	10
Mass-ratio column, sliding	30000	198	260	908	818	238	513
Mass-ratio column, opening	30000	35	36	1878	376	47	36

Scene	Cap [iterations]	Grid reference [iterations]	Fixed structure [iterations]	He, mean [iterations]	He, structure [iterations]	Normalized, mean [iterations]	Normalized, structure [iterations]
Asymmetric tower	30000	2	25811	206	103	41	18
Box lattice	30000	3	68	17	9	30000 C	9
Random contact network	20000	281	354	284	354	284	354
Granular step	20000	1672	20000 C	20000 C	20000 C	20000 C	20000 C
Packing, 50 grains	30000	1180	30000 C	30000 C	30000 C	30000 C	1378
Packing, 200 grains	20000	1398	20000 C	20000 C	20000 C	20000 C	20000 C
Packing, 800 grains	2000	1499	2000 C	2000 C	2000 C	2000 C	2000 C
Dense random contact system	20000	1105	3017	20000 C	20000 C	20000 C	20000 C
Column, heavy top	30000	3282	30000 C	690	1858	955	1373
Eight-body constrained column	30000	4813	30000 C	5103	5762	30000 C	4494
Quadruped, sticking	30000	2	135	13	9	13	9
Quadruped, sliding	30000	807	1020	683	695	1102	1386
Humanoid	30000	3	80	12	11	12	11
Quadruped, unloading probe	30000	92	156	330	302	195	253
Quadruped, batch sticking state	30000	278	293	195	249	1254	1146
Quadruped, batch sliding state	30000	311	501	464	443	1382	773

$R_{\eta} < 10^{-10}$; $N \setminus s, \text{quad } \tau=2, \text{quad } \kappa=10,$
 $\|r_p\| > \kappa \|r_d\| \Rightarrow p \leftarrow \tau p, \text{quad } \|r_d\| > \kappa \|r_p\| \Rightarrow p \leftarrow p/\tau.$

The inner block holds the current tangential-speed vector fixed and refreshes it after the primal residual falls below the following numerical threshold:

$s^{(0)}=0, \quad s_c^{(j+1)} = \|(G_{\eta}z+b)_{\{t,c\}}\|_2,$
 $\delta_0 = \max(\varepsilon, 0.1 \|b\|_{\infty}),$
 $\delta_j = \max(\varepsilon, 0.1 \max(\|s^{(j)} - s^{(j-1)}\|_{\infty}, \varepsilon)), \quad j > 0,$
 $\|x-z\|_2 < \delta_j, \quad \text{total inner updates} \geq 5, \quad \varepsilon = 10^{-10}.$

The threshold compares numerical values in the stated SI convention; it is not a dimensionally invariant criterion. The total-update condition is cumulative across blocks. The corrected natural-map residual is checked at every inner update, before adaptation or shift refresh, and remains the termination criterion. The unscaled dual variable is preserved across penalty changes. These

details distinguish this measured nested iteration from the per-iteration refresh used in the local proposition.

Appendix G. Convex and corrected contact laws

Residuals belong to each method’s own law; the normal-gap diagnostic includes separating contacts.

APGD

APGD can converge to its relaxed-law target while retaining a positive normal-gap diagnostic.

Scene	Iterations [count]	Cap [count]	Status	Residual [Ns], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	101	5000	converged	6.43026743e-11	3.270000014e-05	0
Eight-box tower	2562	5000	converged	9.777877119e-11	2.240763651e-12	0
Mass-ratio column, rest	5000	5000	capped	0.06362969869	2.651237446e-07	0
Mass-ratio column, sliding	5000	5000	capped	0.05286465984	0.0004161165714	0
Mass-ratio column, opening	5000	5000	capped	0.0001746373258	0.0004169794296	0
Asymmetric tower	5000	5000	capped	2.627224926e-05	3.281364609e-08	0
Box lattice	576	5000	converged	6.206488448e-11	5.4977753e-20	0
Random contact network	3079	5000	converged	8.262893147e-11	0.001516901924	0
Granular step	5000	5000	capped	9.343661489e-09	0.01291150628	0
Packing, 50 grains	3895	5000	converged	9.950401516e-11	0.005844353264	0
Packing, 200 grains	5000	5000	capped	3.011538213e-10	0.005998695167	0
Packing, 800 grains	5000	5000	capped	9.46212492e-10	0.005989463739	0
Dense random contact system	1221	5000	converged	9.775151522e-11	1.35097248e-10	0
Column, heavy top	5000	5000	capped	0.07635563843	0.0008333580174	0
Eight-body constrained column	5000	5000	capped	0.4616320989	0.0008334576305	0
Quadruped, sticking	468	5000	converged	9.738644613e-11	2.266645089e-12	0
Quadruped, sliding	817	5000	converged	2.782746256e-11	5.286511295e-06	0
Humanoid	229	5000	converged	6.667943975e-11	0	0
Quadruped, unloading probe	402	5000	converged	6.5659575e-11	8.001231629e-05	0

Scene	Iterations [count]	Cap [count]	Status	Residual [Ns], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Quadruped, batch sticking state	842	5000	converged	7.506481547e-11	3.733374691e-06	0
Quadruped, batch sliding state	818	5000	converged	3.688355177e-12	1.223284581e-05	0

Convex PGS

The scene-dependent budgets leave several projected Gauss-Seidel runs above their relaxed-law target.

Scene	Iterations [count]	Cap [count]	Status	Residual [Ns], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	53	5000	converged	8.949960858e-11	3.270000003e-05	0
Eight-box tower	4500	4500	capped	2.519047193e-06	5.772816483e-08	0
Mass-ratio column, rest	5000	5000	capped	0.006788612785	0	0
Mass-ratio column, sliding	5000	5000	capped	0.00678513913	0.000416032746	0
Mass-ratio column, opening	5000	5000	capped	4.562734586e-05	0.0004178030123	0
Asymmetric tower	4500	4500	capped	0.002165968144	0	0
Box lattice	261	5000	converged	9.881346782e-11	5.314010724e-12	0
Random contact network	644	1000	converged	9.923387112e-11	0.001516901924	0
Granular step	400	400	capped	1.47431094e-05	0.01291050442	0
Packing, 50 grains	913	1000	converged	9.886788437e-11	0.005844353266	0
Packing, 200 grains	250	250	capped	1.347454571e-06	0.005998770029	0
Packing, 800 grains	60	60	capped	3.315655711e-05	0.00598916608	0
Dense random contact system	550	550	capped	6.715880001e-06	4.058504171e-05	0
Column, heavy top	5000	5000	capped	0.008146335342	0.0008333333333	0
Eight-body constrained column	5000	5000	capped	0.04037576716	0.0008314102311	0
Quadruped, sticking	2490	5000	converged	9.983275318e-11	2.281182332e-12	0
Quadruped, sliding	1341	5000	converged	9.984302274e-11	5.286507372e-06	0
Humanoid	198	5000	converged	9.80975301e-11	0	0

Scene	Iterations [count]	Cap [count]	Status	Residual [Ns], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Quadruped, unloading probe	418	5000	converged	9.713191362e-11	8.001230962e-05	0
Quadruped, batch sticking state	1310	5000	converged	9.851050531e-11	3.733367731e-06	0
Quadruped, batch sliding state	1146	5000	converged	9.879524976e-11	1.22328407e-05	0

Coulomb ADMM

The corrected solver reaches its target in every listed scene; positive normal gaps remain possible at separating contacts.

Scene	Iterations [count]	Cap [count]	Status	Residual [Ns], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Cube	70	5000	converged	7.450456918e-11	1.422830892e-13	0
Eight-box tower	2	5000	converged	4.951733468e-13	0	0
Mass-ratio column, rest	5	5000	converged	2.273736754e-13	4.736951572e-16	0
Mass-ratio column, sliding	198	5000	converged	8.446932043e-11	4.736951572e-16	0
Mass-ratio column, opening	35	5000	converged	7.668177204e-11	0.0004177078125	0
Asymmetric tower	2	5000	converged	4.170885859e-12	8.21037104e-14	0
Box lattice	3	5000	converged	1.085208312e-13	1.275469654e-15	0
Random contact network	281	5000	converged	9.019551425e-11	0.001688735757	0
Granular step	1672	5000	converged	9.982629307e-11	0.01320401497	0
Packing, 50 grains	1180	5000	converged	9.947796102e-11	0.005844240581	0
Packing, 200 grains	1398	5000	converged	9.938339279e-11	0.006096702174	0
Packing, 800 grains	1499	5000	converged	9.985446237e-11	0.005989526298	0
Dense random contact system	1105	5000	converged	9.991044971e-11	0.02370284443	0
Column, heavy top	3282	5000	converged	9.958966984e-11	0	0
Eight-body constrained column	4813	5000	converged	9.965783754e-11	0	0
Quadruped, sticking	2	5000	converged	2.059552529e-11	0	0
Quadruped, sliding	807	5000	converged	8.900271709e-11	2.582387891e-12	0

Scene	Iterations [count]	Cap [count]	Status	Residual [N s], target $1e-10$	Normal gap [m]	Tangential test failures [1]
Humanoid	3	5000	converged	6.562261845e-11	0	0
Quadruped, unloading probe	92	5000	converged	2.403774488e-11	0	0
Quadruped, batch sticking state	278	5000	converged	9.285461549e-11	2.285389181e-13	0
Quadruped, batch sliding state	311	5000	converged	6.128820368e-11	0	0

$\Delta t = 1/240$; s, quad \text{tangential test fails if } $\|\lambda_t\| > \mu \lambda_n + 10^{-12}$; N, s.

Appendix H. External friction-contact solutions

Siconos relative stopping errors and absolute natural-map residuals are shown separately, with no matched-accuracy speed inference.

NSGS

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	28	3.119242636e-11	7.080810072e-12	0.05590169703
Eight-box tower	converged	36138	9.994128621e-11	5.98493477e-13	0.2220458214
Mass-ratio column, rest	converged	23061	9.981710826e-11	9.993073036e-11	9.917666169e-11
Mass-ratio column, sliding	converged	22526	8.510786222e-11	2.252136255e-10	2.050074307e-10
Mass-ratio column, opening	converged	11443	9.987405774e-11	1.163584784e-10	5.353683565e-09
Asymmetric tower	iteration cap	200000	8.063739006e-05	1.667788212e-05	0.5048144963
Box lattice	converged	201	9.7574752e-11	3.969533036e-13	0.4111027742
Random contact network	converged	122	2.862340751e-11	2.352779083e-11	4.249420871e-07
Granular step	converged	482	9.909073602e-11	1.632661194e-11	4.699232181e-06
Packing, 50 grains	converged	557	2.747599854e-11	6.188151472e-14	0.0008072604959
Packing, 200 grains	converged	696	4.811111969e-11	4.975932318e-13	0.0001995416797
Packing, 800 grains	converged	724	9.043927436e-11	5.928003314e-13	7.287681841e-05
Dense random contact system	converged	576	6.625714818e-11	8.813059657e-12	4.147401393e-06
Column, heavy top	converged	22602	9.994367934e-11	1.265334504e-10	1.241471842e-10
Eight-body constrained column	converged	73028	5.56297353e-11	1.045660225e-10	4.429070464e-07
Quadruped, sticking	converged	132	4.870892392e-11	4.181724411e-13	2.906827478e-09

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Quadruped, sliding	converged	92	1.267880645e-11	4.226619055e-13	8.52151708e-06
Humanoid	converged	6	9.815478326e-14	8.205242041e-15	9.958433509e-10
Quadruped, unloading probe	converged	6	2.518981392e-15	3.122502257e-17	1.872049997e-06
Quadruped, batch sticking state	converged	43	3.093408589e-11	3.750055821e-13	6.215054932e-07
Quadruped, batch sliding state	converged	71	8.273748574e-12	2.959195389e-13	2.735310156e-06

ACLMFP

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	66	8.782108299e-11	1.132571814e-11	0.1218080142
Eight-box tower	converged	47408	9.998550962e-11	5.463268726e-13	0.2214364408
Mass-ratio column, rest	converged	29471	9.998225769e-11	6.069490155e-08	1.856154512e-10
Mass-ratio column, sliding	converged	17497	9.99801098e-11	9.989616956e-08	4.195114329e-10
Mass-ratio column, opening	converged	10846	9.998990048e-11	1.081730261e-10	5.388817994e-09
Asymmetric tower	iteration cap	100000	6.61512121e-05	1.15881481e-05	0.4946032172
Box lattice	converged	257	9.873123625e-11	3.368407983e-13	0.4071982673
Random contact network	converged	158	9.331661175e-11	3.662745801e-11	4.266287422e-07
Granular step	converged	676	9.967782436e-11	2.162775167e-11	4.690862821e-06
Packing, 50 grains	iteration cap	100000	2.434586967e-08	5.779904724e-11	0.0008068265272
Packing, 200 grains	wall limit or error	—	—	—	—
Packing, 800 grains	wall limit or error	—	—	—	—
Dense random contact system	wall limit or error	—	—	—	—
Column, heavy top	converged	24805	9.996339867e-11	7.550249848e-08	2.464333506e-10
Eight-body constrained column	converged	85977	9.999657395e-11	1.073743594e-07	1.563755988e-06
Quadruped, sticking	converged	1068	9.958796911e-11	7.861489237e-13	3.155308899e-09
Quadruped, sliding	converged	687	9.815923189e-11	2.751368577e-12	8.519220228e-06
Humanoid	converged	141	9.949239809e-11	7.692957382e-12	8.919651816e-10
Quadruped, unloading probe	converged	31	7.646929485e-11	5.260653024e-13	1.872087784e-06
Quadruped, batch sticking state	converged	233	9.556573987e-11	1.290204055e-12	6.212439137e-07

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Quadruped, batch sliding state	converged	605	9.532804496e-11	3.17216392e-12	2.732522416e-06

PROX

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	21	1.195048333e-15	2.498001805e-16	5.052887144e-08
Eight-box tower	converged	26	1.187564286e-11	4.485301019e-14	1.000224453e-06
Mass-ratio column, rest	converged	8910	1.72414626e-12	1.317857823e-09	2.941058437e-11
Mass-ratio column, sliding	converged	4567	6.669023245e-12	7.414200809e-09	6.309629157e-11
Mass-ratio column, opening	converged	3044	5.680060299e-11	4.234834705e-11	5.322042327e-09
Asymmetric tower	converged	443	8.064423322e-13	1.207922651e-13	0.011978239
Box lattice	converged	17	7.483188261e-15	3.304411454e-16	2.943114736e-07
Random contact network	converged	7	3.536199358e-16	2.220444801e-16	4.251110231e-07
Granular step	converged	4	9.179025474e-14	6.537218683e-14	4.696746953e-06
Packing, 50 grains	converged	59	3.816544388e-12	2.902707998e-13	0.0008072591948
Packing, 200 grains	wall limit or error	—	—	—	—
Packing, 800 grains	wall limit or error	—	—	—	—
Dense random contact system	converged	5	3.817844118e-12	3.80141179e-11	4.145632619e-06
Column, heavy top	converged	5640	1.271057486e-13	2.155502443e-10	3.220425029e-11
Eight-body constrained column	converged	22	1.710280014e-11	2.192547377e-08	5.364301665e-07
Quadruped, sticking	converged	22	1.1288416e-12	8.54871729e-15	2.90556666e-09
Quadruped, sliding	converged	14	3.736126891e-15	2.892651396e-16	8.521804576e-06
Humanoid	converged	92	6.001882487e-17	8.881784197e-16	9.959136974e-10
Quadruped, unloading probe	converged	15	8.235519967e-16	1.387778781e-17	1.872049998e-06
Quadruped, batch sticking state	converged	25	4.052399307e-11	5.354050536e-13	6.214240505e-07
Quadruped, batch sliding state	converged	11	3.391772795e-11	2.072163188e-12	2.734630354e-06

ADMM

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	100	1.99962858e-12	3.853861674e-13	5.052298146e-08
Eight-box tower	converged	43	8.838426188e-11	3.071015664e-13	3.212830341e-10
Mass-ratio column, rest	converged	15213	3.351371511e-11	2.376748398e-08	6.695662359e-11
Mass-ratio column, sliding	converged	145713	5.492617119e-11	6.31102921e-08	2.54125747e-10
Mass-ratio column, opening	converged	465	9.554086874e-11	7.878497854e-11	5.39477999e-09
Asymmetric tower	iteration cap	200000	2.233042998e-05	2.479208142e-06	0.07748569217
Box lattice	converged	26	5.146269863e-12	3.123890036e-14	6.895668638e-08
Random contact network	converged	231	2.025274095e-11	3.128269573e-11	4.250891628e-07
Granular step	converged	1339	1.328375051e-11	2.005329609e-11	4.696756437e-06
Packing, 50 grains	converged	3478	2.810092043e-11	9.556434207e-12	0.0008072571709
Packing, 200 grains	converged	4785	1.603306024e-12	8.371752456e-13	0.000199541461
Packing, 800 grains	converged	4585	1.282413998e-11	1.122244946e-11	7.287686058e-05
Dense random contact system	wall limit or error	—	—	—	—
Column, heavy top	converged	101083	9.996463504e-11	8.752579106e-08	2.693286516e-10
Eight-body constrained column	iteration cap	200000	2.486150043e-08	3.351768052e-05	0.0003651861153
Quadruped, sticking	converged	125	3.584456863e-12	2.57294186e-14	2.900985165e-09
Quadruped, sliding	converged	1055	5.972642016e-11	2.404035304e-12	8.520896583e-06
Humanoid	converged	113	5.799374575e-12	4.151123889e-13	9.898142908e-10
Quadruped, unloading probe	converged	172	2.008777701e-12	5.851569229e-14	1.872055084e-06
Quadruped, batch sticking state	converged	224	3.886356208e-12	1.293635338e-13	6.215954419e-07
Quadruped, batch sliding state	converged	1152	1.944582277e-15	3.365363543e-16	2.735523256e-06

DSFP

The de Saxcé fixed-point method includes nonfinite outputs and capped runs; these are failures, irrespective of the reported iteration count.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	97	8.885339433e-11	1.667976868e-11	5.065120362e-08
Eight-box tower	iteration cap	200000	2.82236756e+87	1.118521162e+85	7.852524632e+85
Mass-ratio column, rest	iteration cap	200000	1.013473563e+21	7.163376343e+20	4.477921096e+17
Mass-ratio column, sliding	iteration cap	200000	71.29839568	90.83869789	0.919838375
Mass-ratio column, opening	iteration cap	200000	4.73494208e+20	3.902918378e+20	5.9761805e+17

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Asymmetric tower	iteration cap	200000	4.320894993e+153	3.323632051e+153	5.589988934e+151
Box lattice	iteration cap	200000	580.2490844	7.783789091	251.1975491
Random contact network	iteration cap	200000	nonfinite	—	nonfinite
Granular step	iteration cap	200000	nonfinite	—	nonfinite
Packing, 50 grains	iteration cap	200000	2.02350991e+105	1.251296802e+103	8.202543881e+105
Packing, 200 grains	iteration cap	200000	5.245513264e+38	4.477906324e+36	1.078227256e+39
Packing, 800 grains	wall limit or error	—	—	—	—
Dense random contact system	wall limit or error	—	—	—	—
Column, heavy top	converged	93380	9.998800196e-11	8.950667052e-08	2.773120989e-10
Eight-body constrained column	iteration cap	200000	2.527415886e-05	5.135840246e-05	0.0006414031528
Quadruped, sticking	iteration cap	200000	1.158875492	0.02391664236	0.6543405324
Quadruped, sliding	iteration cap	200000	0.33817247	0.02534091624	0.7600255177
Humanoid	converged	405	9.971855857e-11	8.047562616e-12	8.92613923e-10
Quadruped, unloading probe	iteration cap	200000	1	0.005372673136	1
Quadruped, batch sticking state	iteration cap	200000	0.9465417767	0.03907821554	0.7399587151
Quadruped, batch sliding state	iteration cap	200000	0.2633533011	0.02668787057	0.7235440118

NSN AC

The statuses and attained residuals must be read together: the Siconos stopping test does not impose the absolute natural-map target.

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Cube	converged	1	2.502018505e-16	1.592316813e-17	0.4472135764
Eight-box tower	converged	730	1.957576129e-12	2.011585343e-14	1.131419213
Mass-ratio column, rest	converged	3	1.607774678e-13	2.273736754e-13	1.128825108e-12
Mass-ratio column, sliding	converged	4	1.200666826e-13	1.705302566e-13	2.565752091e-11
Mass-ratio column, opening	converged	2	9.748565909e-14	1.136868377e-13	5.237091862e-09
Asymmetric tower	iteration cap	10001	0.01973507073	0.006133655368	0.8306568771
Box lattice	converged	2814	1.266132896e-11	8.619702174e-13	1.190006091
Random contact network	converged	7	4.354035828e-11	1.031931103e-10	4.25157156e-07
Granular step	converged	13	7.220457066e-15	3.229447959e-14	4.696747562e-06
Packing, 50 grains	converged	14	1.984925267e-11	5.188776756e-12	0.000807261616

Scene	Status	Iterations [count]	Siconos residual [1], target $1e-10$	Natural-map residual [N s]	Relative impulse difference [1]
Packing, 200 grains	wall limit or error	—	—	—	—
Packing, 800 grains	wall limit or error	—	—	—	—
Dense random contact system	converged	14	2.299109643e-11	1.399806469e-10	4.147557684e-06
Column, heavy top	converged	3	1.44922851e-13	2.273736754e-13	3.477711282e-11
Eight-body constrained column	converged	26	1.812986607e-16	2.442490654e-15	7.221080274e-07
Quadruped, sticking	converged	1	6.656433011e-16	6.938893904e-18	2.907839753e-09
Quadruped, sliding	converged	4	1.230955805e-11	1.894063899e-12	8.521901905e-06
Humanoid	converged	1	2.400752995e-16	2.775557562e-17	9.959044269e-10
Quadruped, unloading probe	converged	4	2.541362653e-13	2.102484853e-15	1.872049774e-06
Quadruped, batch sticking state	converged	6	3.709804386e-16	1.734723476e-18	6.215954638e-07
Quadruped, batch sliding state	converged	5	4.35430705e-12	5.480802444e-13	2.735553634e-06

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